

Counting Sample Points

2.21 Registrants at a large convention are offered 6 sightseeing tours on each of 3 days. In how many ways can a person arrange to go on a sightseeing tour planned by this convention?

2.22 In a medical study, patients are classified in 8 ways according to whether they have blood type AB^+ , AB^- , A^+ , A^- , B^+ , B^- , O^+ , or O^- , and also according to whether their blood pressure is low, normal, or high. Find the number of ways in which a patient can be classified.

2.23 If an experiment consists of throwing a die and then drawing a letter at random from the English alphabet, how many points are there in the sample space?

2.24 Students at a private liberal arts college are classified as being freshmen, sophomores, juniors, or seniors, and also according to whether they are male or female. Find the total number of possible classifications for the students of that college.

2.25 A certain brand of shoes comes in 5 different styles, with each style available in 4 distinct colors. If the store wishes to display pairs of these shoes showing all of its various styles and colors, how many different pairs will the store have on display?

2.26 A California study concluded that following 7 simple health rules can extend a man's life by 11 years on the average and a woman's life by 7 years. These 7 rules are as follows: no smoking, get regular exercise, use alcohol only in moderation, get 7 to 8 hours of sleep, maintain proper weight, eat breakfast, and do

not eat between meals. In how many ways can a person adopt 5 of these rules to follow

- (a) if the person presently violates all 7 rules?
- (b) if the person never drinks and always eats breakfast?

2.27 A developer of a new subdivision offers a prospective home buyer a choice of 4 designs, 3 different heating systems, a garage or carport, and a patio or screened porch. How many different plans are available to this buyer?

2.28 A drug for the relief of asthma can be purchased from 5 different manufacturers in liquid, tablet, or capsule form, all of which come in regular and extra strength. How many different ways can a doctor prescribe the drug for a patient suffering from asthma?

2.29 In a fuel economy study, each of 3 race cars is tested using 5 different brands of gasoline at 7 test sites located in different regions of the country. If 2 drivers are used in the study, and test runs are made once under each distinct set of conditions, how many test runs are needed?

2.30 In how many different ways can a true-false test consisting of 9 questions be answered?

2.31 A witness to a hit-and-run accident told the police that the license number contained the letters RLH followed by 3 digits, the first of which was a 5. If the witness cannot recall the last 2 digits, but is certain that all 3 digits are different, find the maximum number of automobile registrations that the police may have to check.

- 2.32** (a) In how many ways can 6 people be lined up to get on a bus?
 (b) If 3 specific persons, among 6, insist on following each other, how many ways are possible?
 (c) If 2 specific persons, among 6, refuse to follow each other, how many ways are possible?
- 2.33** If a multiple-choice test consists of 5 questions, each with 4 possible answers of which only 1 is correct,
 (a) in how many different ways can a student check off one answer to each question?
 (b) in how many ways can a student check off one answer to each question and get all the answers wrong?
- 2.34** (a) How many distinct permutations can be made from the letters of the word *COLUMNS*?
 (b) How many of these permutations start with the letter *M*?
- 2.35** A contractor wishes to build 9 houses, each different in design. In how many ways can he place these houses on a street if 6 lots are on one side of the street and 3 lots are on the opposite side?
- 2.36** (a) How many three-digit numbers can be formed from the digits 0, 1, 2, 3, 4, 5, and 6 if each digit can be used only once?
 (b) How many of these are odd numbers?
 (c) How many are greater than 330?
- 2.37** In how many ways can 4 boys and 5 girls sit in a row if the boys and girls must alternate?
- 2.38** Four married couples have bought 8 seats in the same row for a concert. In how many different ways can they be seated
- (c) if all the men sit together to the right of all the women?
- 2.39** In a regional spelling bee, the 8 finalists consist of 3 boys and 5 girls. Find the number of sample points in the sample space S for the number of possible orders at the conclusion of the contest for
 (a) all 8 finalists;
 (b) the first 3 positions.
- 2.40** In how many ways can 5 starting positions on a basketball team be filled with 8 men who can play any of the positions?
- 2.41** Find the number of ways that 6 teachers can be assigned to 4 sections of an introductory psychology course if no teacher is assigned to more than one section.
- 2.42** Three lottery tickets for first, second, and third prizes are drawn from a group of 40 tickets. Find the number of sample points in S for awarding the 3 prizes if each contestant holds only 1 ticket.
- 2.43** In how many ways can 5 different trees be planted in a circle?
- 2.44** In how many ways can a caravan of 8 covered wagons from Arizona be arranged in a circle?
- 2.45** How many distinct permutations can be made from the letters of the word *INFINITY*?
- 2.46** In how many ways can 3 oaks, 4 pines, and 2 maples be arranged along a property line if one does not distinguish among trees of the same kind?
- 2.47** How many ways are there to select 3 candidates from 8 equally qualified recent graduates for openings in an accounting firm?
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